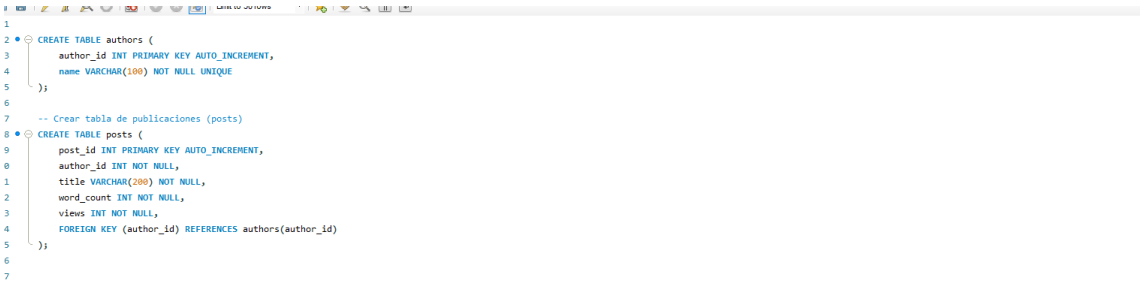
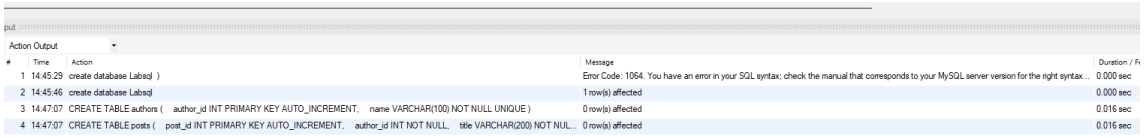
1. Normalize the following blog database and write the DDL scripts to create the database tables:



```sql
CREATE TABLE authors (
    author_id INT PRIMARY KEY AUTO_INCREMENT,
    name VARCHAR(100) NOT NULL UNIQUE
);

-- Crear tabla de publicaciones (posts)
CREATE TABLE posts (
    post_id INT PRIMARY KEY AUTO_INCREMENT,
    author_id INT NOT NULL,
    title VARCHAR(200) NOT NULL,
    word_count INT NOT NULL,
    views INT NOT NULL,
    FOREIGN KEY (author_id) REFERENCES authors(author_id)
);
```



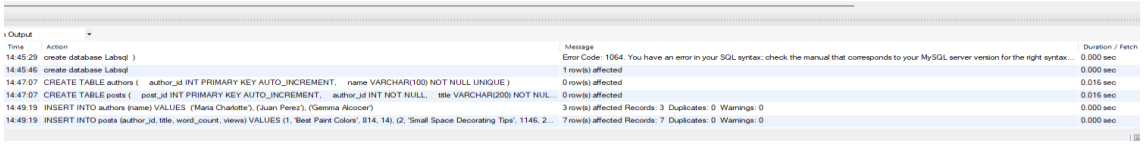
Action Output

#	Time	Action	Message	Duration / Fe
1	14:45:29	create database Labsql)	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax...	0.000 sec
2	14:45:46	create database Labsql	1 row(s) affected	0.000 sec
3	14:47:07	CREATE TABLE authors (author_id INT PRIMARY KEY AUTO_INCREMENT, name VARCHAR(100) NOT NULL UNIQUE)	0 row(s) affected	0.016 sec
4	14:47:07	CREATE TABLE posts (post_id INT PRIMARY KEY AUTO_INCREMENT, author_id INT NOT NULL, title VARCHAR(200) NOT NUL...	0 row(s) affected	0.016 sec



```sql
-- Insertar autores
INSERT INTO authors (name) VALUES
('Maria Charlotte'),
('Juan Perez'),
('Gemma Alcocer');

-- Insertar publicaciones
INSERT INTO posts (author_id, title, word_count, views) VALUES
(1, 'Best Paint Colors', 814, 14),
(2, 'Small Space Decorating Tips', 1146, 221),
(1, 'Hot Accessories', 986, 105),
(1, 'Mixing Textures', 765, 22),
(2, 'Kitchen Refresh', 1242, 307),
(1, 'Homemade Art Hacks', 1002, 193),
(3, 'Refinishing Wood Floors', 1571, 7542);
```



Output

Time	Action	Message	Duration / Fetch
14:45:29	create database Labsql)	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax...	0.000 sec
14:45:46	create database Labsql	1 row(s) affected	0.000 sec
14:47:07	CREATE TABLE authors (author_id INT PRIMARY KEY AUTO_INCREMENT, name VARCHAR(100) NOT NULL UNIQUE)	0 row(s) affected	0.016 sec
14:47:07	CREATE TABLE posts (post_id INT PRIMARY KEY AUTO_INCREMENT, author_id INT NOT NULL, title VARCHAR(200) NOT NUL...	0 row(s) affected	0.016 sec
14:49:19	INSERT INTO authors (name) VALUES ('Maria Charlotte'), ('Juan Perez'), ('Gemma Alcocer')	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.000 sec
14:49:19	INSERT INTO posts (author_id, title, word_count, views) VALUES (1, 'Best Paint Colors', 814, 14), (2, 'Small Space Decorating Tips', 1146, 2...	7 row(s) affected Records: 7 Duplicates: 0 Warnings: 0	0.000 sec

2. Normalize the following airline database and write the DDL scripts to create the database tables:

```sql
-- Tabla de clientes
CREATE TABLE customers (
    customer_id INT PRIMARY KEY AUTO_INCREMENT,
    name VARCHAR(100) NOT NULL,
    status VARCHAR(20),
    total_mileage INT,
    UNIQUE(name)  -- Asumiendo que el nombre es único para simplificar
);
```

Action Output

	Time	Action	Message	Duration / Fetch
2	14:45:46	create database Labsql	1 row(s) affected	0.000 sec
3	14:47:07	CREATE TABLE authors (author_id INT PRIMARY KEY AUTO_INCREMENT, name VARCHAR(100) NOT NULL UNIQUE)	0 row(s) affected	0.016 sec
4	14:47:07	CREATE TABLE posts (post_id INT PRIMARY KEY AUTO_INCREMENT, author_id INT NOT NULL, title VARCHAR(200) NOT N...	0 row(s) affected	0.016 sec
5	14:49:19	INSERT INTO authors (name) VALUES ('Maria Charlotte'), ('Juan Perez'), ('Gemma Alcocer')	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.000 sec
6	14:49:19	INSERT INTO posts (author_id, title, word_count, views) VALUES (1, 'Best Paint Colors', 814, 14), (2, 'Small Space Decorating Tips', 1146, ...	7 row(s) affected Records: 7 Duplicates: 0 Warnings: 0	0.000 sec
7	14:54:35	CREATE TABLE customers (customer_id INT PRIMARY KEY AUTO_INCREMENT, name VARCHAR(100) NOT NULL, status VA...	0 row(s) affected	0.015 sec

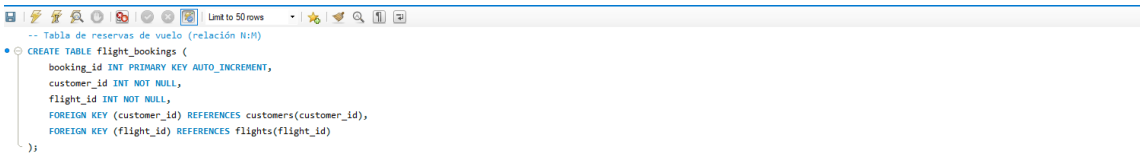
```sql
-- Tabla de aeronaves
CREATE TABLE aircrafts (
    aircraft_id INT PRIMARY KEY AUTO_INCREMENT,
    model VARCHAR(50) NOT NULL UNIQUE,
    total_seats INT NOT NULL
);
```

Action Output

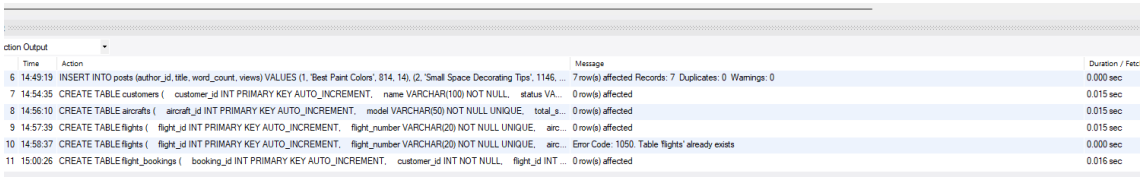
	Time	Action	Message	Duration / Fetch
3	14:47:07	CREATE TABLE authors (author_id INT PRIMARY KEY AUTO_INCREMENT, name VARCHAR(100) NOT NULL UNIQUE)	0 row(s) affected	0.016 sec
4	14:47:07	CREATE TABLE posts (post_id INT PRIMARY KEY AUTO_INCREMENT, author_id INT NOT NULL, title VARCHAR(200) NOT N...	0 row(s) affected	0.016 sec
5	14:49:19	INSERT INTO authors (name) VALUES ('Maria Charlotte'), ('Juan Perez'), ('Gemma Alcocer')	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.000 sec
6	14:49:19	INSERT INTO posts (author_id, title, word_count, views) VALUES (1, 'Best Paint Colors', 814, 14), (2, 'Small Space Decorating Tips', 1146, ...	7 row(s) affected Records: 7 Duplicates: 0 Warnings: 0	0.000 sec
7	14:54:35	CREATE TABLE customers (customer_id INT PRIMARY KEY AUTO_INCREMENT, name VARCHAR(100) NOT NULL, status VA...	0 row(s) affected	0.015 sec
8	14:56:10	CREATE TABLE aircrafts (aircraft_id INT PRIMARY KEY AUTO_INCREMENT, model VARCHAR(50) NOT NULL UNIQUE, total_s...	0 row(s) affected	0.015 sec

```sql
-- Tabla de vuelos
CREATE TABLE flights (
    flight_id INT PRIMARY KEY AUTO_INCREMENT,
    flight_number VARCHAR(20) NOT NULL UNIQUE,
    aircraft_id INT NOT NULL,
    mileage INT NOT NULL,
    FOREIGN KEY (aircraft_id) REFERENCES aircrafts(aircraft_id)
);
```

Time	Action	Message	Duration /
14:49:19	INSERT INTO authors (name) VALUES ('Maria Charlotte'), ('Juan Perez'), ('Gemma Alcocer')	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.000 sec
14:49:19	INSERT INTO posts (author_id, title, word_count, views) VALUES (1, 'Best Paint Colors', 814, 14), (2, 'Small Space Decorating Tips', 1146, ...	7 row(s) affected Records: 7 Duplicates: 0 Warnings: 0	0.000 sec
14:54:35	CREATE TABLE customers (customer_id INT PRIMARY KEY AUTO_INCREMENT, name VARCHAR(100) NOT NULL, status VA...	0 row(s) affected	0.015 sec
14:56:10	CREATE TABLE aircrafts (aircraft_id INT PRIMARY KEY AUTO_INCREMENT, model VARCHAR(50) NOT NULL UNIQUE, total_s...	0 row(s) affected	0.015 sec
14:57:39	CREATE TABLE flights (flight_id INT PRIMARY KEY AUTO_INCREMENT, flight_number VARCHAR(20) NOT NULL UNIQUE, airc...	0 row(s) affected	0.015 sec



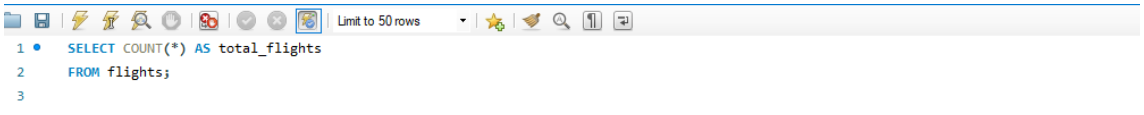
```
-- Tabla de reservas de vuelo (relación N:M)
CREATE TABLE flight_bookings (
    booking_id INT PRIMARY KEY AUTO_INCREMENT,
    customer_id INT NOT NULL,
    flight_id INT NOT NULL,
    FOREIGN KEY (customer_id) REFERENCES customers(customer_id),
    FOREIGN KEY (flight_id) REFERENCES flights(flight_id)
);
```



ction Output

	Time	Action	Message	Duration / Fetc
6	14:49:19	INSERT INTO posts (author_id, title, word_count, views) VALUES (1, 'Best Paint Colors', 814, 14), (2, 'Small Space Decorating Tips', 1146, ...	7 row(s) affected Records: 7 Duplicates: 0 Warnings: 0	0.000 sec
7	14:54:35	CREATE TABLE customers (customer_id INT PRIMARY KEY AUTO_INCREMENT, name VARCHAR(100) NOT NULL, status VA...	0 row(s) affected	0.015 sec
8	14:56:10	CREATE TABLE aircrafts (aircraft_id INT PRIMARY KEY AUTO_INCREMENT, model VARCHAR(50) NOT NULL UNIQUE, total_s...	0 row(s) affected	0.015 sec
9	14:57:39	CREATE TABLE flights (flight_id INT PRIMARY KEY AUTO_INCREMENT, flight_number VARCHAR(20) NOT NULL UNIQUE, airc...	0 row(s) affected	0.015 sec
10	14:58:37	CREATE TABLE flights (flight_id INT PRIMARY KEY AUTO_INCREMENT, flight_number VARCHAR(20) NOT NULL UNIQUE, airc...	Error Code: 1050. Table 'flights' already exists	0.000 sec
11	15:00:26	CREATE TABLE flight_bookings (booking_id INT PRIMARY KEY AUTO_INCREMENT, customer_id INT NOT NULL, flight_id INT ...	0 row(s) affected	0.016 sec

3.In the Airline database write the SQL script to get the total number of flights in the database.



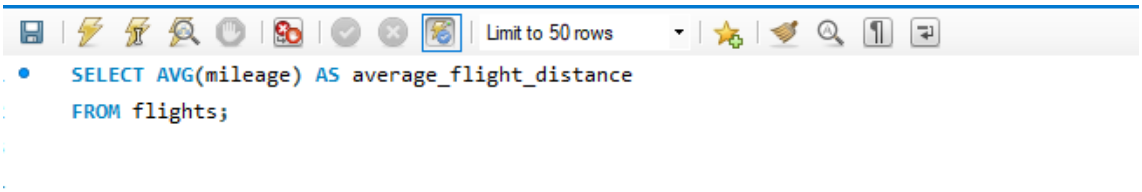
```
SELECT COUNT(*) AS total_flights
FROM flights;
```

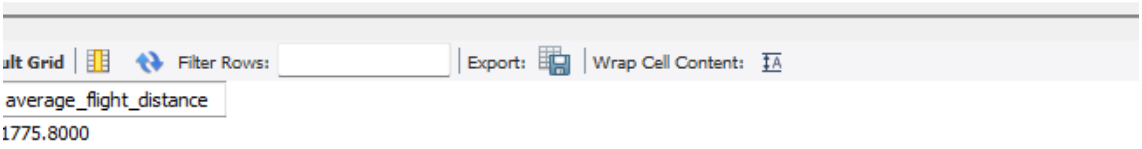


Result Grid | Filter Rows: | Export: | Wrap Cell Content:

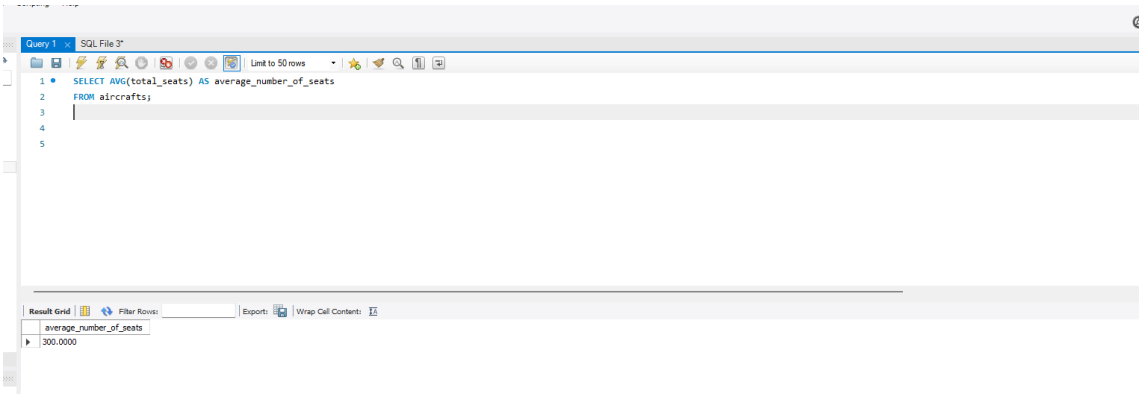
total_flights
5

4. In the Airline database write the SQL script to get the average flight distance.

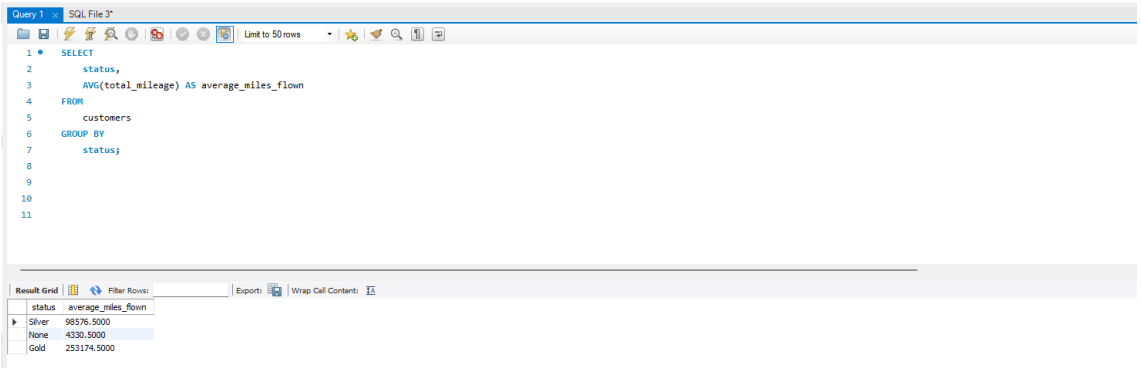




5.In the Airline database write the SQL script to get the average number of seats.



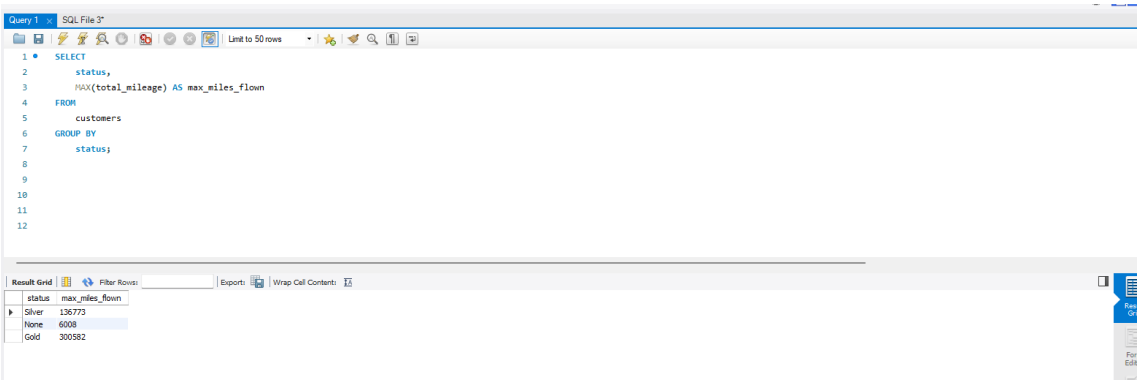
6. In the Airline database write the SQL script to get the average number of miles flown by customers grouped by status.



```
SELECT
    status,
    AVG(total_mileage) AS average_miles_flown
FROM
    customers
GROUP BY
    status;
```

status	average_miles_flown
Silver	98576.5000
None	4330.5000
Gold	253174.5000

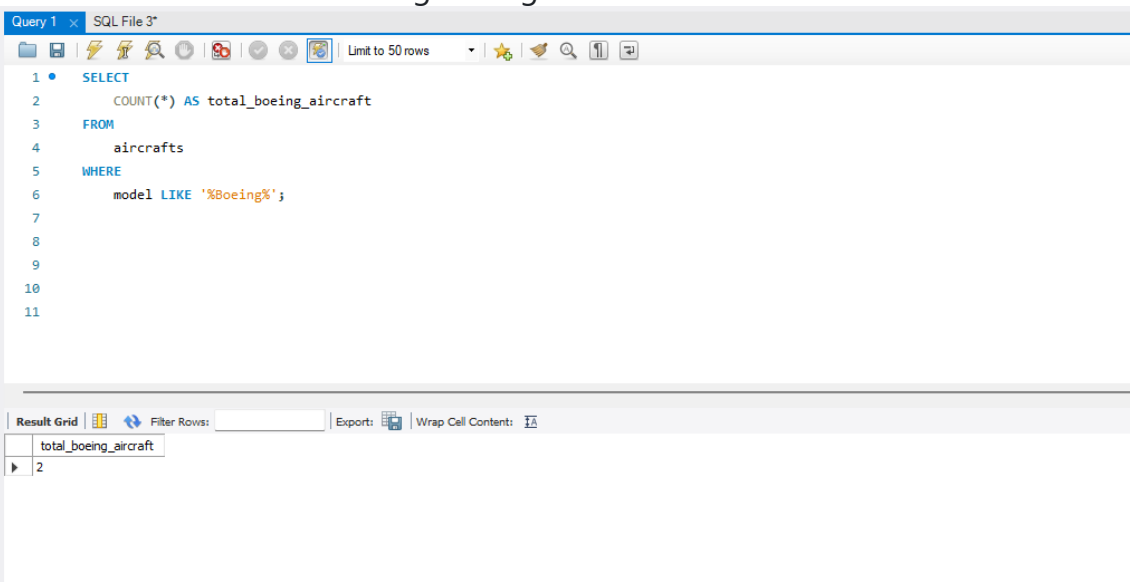
7. In the Airline database write the SQL script to get the maximum number of miles flown by customers grouped by status.



```
SELECT
    status,
    MAX(total_mileage) AS max_miles_flown
FROM
    customers
GROUP BY
    status;
```

status	max_miles_flown
Silver	136773
None	6008
Gold	300582

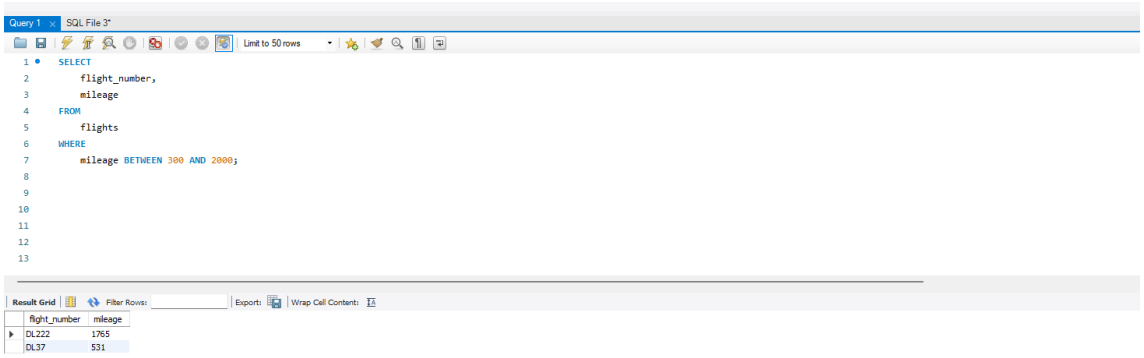
8. In the Airline database write the SQL script to get the total number of aircraft with a name containing Boeing.



```
SELECT
    COUNT(*) AS total_boeing_aircraft
FROM
    aircrafts
WHERE
    model LIKE '%Boeing%';
```

total_boeing_aircraft
2

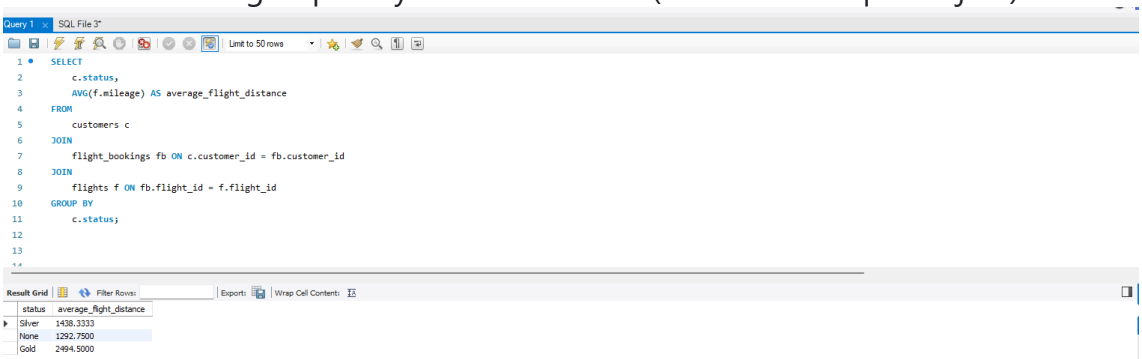
9. In the Airline database write the SQL script to find all flights with a distance between 300 and 2000 miles.



```
SELECT
    flight_number,
    mileage
FROM
    flights
WHERE
    mileage BETWEEN 300 AND 2000;
```

flight_number	mileage
DL222	1765
DL37	531

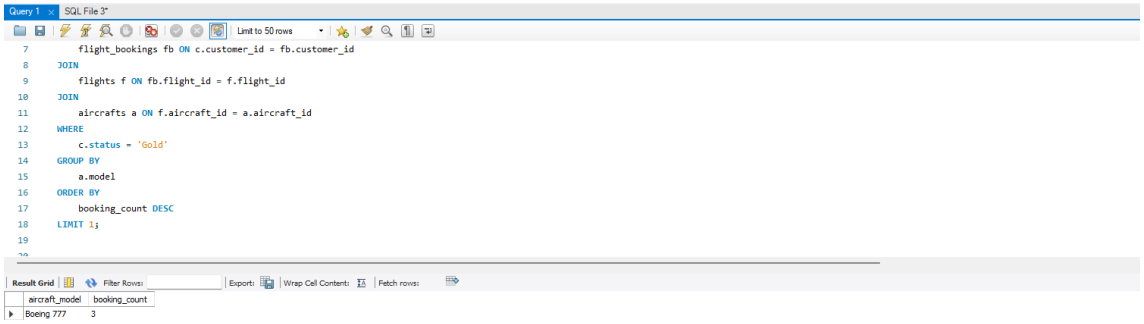
10. In the Airline database write the SQL script to find the average flight distance booked grouped by customer status (this should require a join).



```
SELECT
    c.status,
    AVG(f.mileage) AS average_flight_distance
FROM
    customers c
JOIN
    flight_bookings fb ON c.customer_id = fb.customer_id
JOIN
    flights f ON fb.flight_id = f.flight_id
GROUP BY
    c.status;
```

status	average_flight_distance
Silver	1438.3333
None	1292.7500
Gold	2494.5000

11. In the Airline database write the SQL script to find the most often booked aircraft by gold status members (this should require a join).



```
    flight_bookings fb ON c.customer_id = fb.customer_id
JOIN
    flights f ON fb.flight_id = f.flight_id
JOIN
    aircrafts a ON f.aircraft_id = a.aircraft_id
WHERE
    c.status = 'Gold'
GROUP BY
    a.model
ORDER BY
    booking_count DESC
LIMIT 1;
```

aircraft_model	booking_count
Boeing 777	3